

Project 2: High Fidelity App Design Documentation

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Figma Prototype Link:

<https://www.figma.com/design/4y2igbTD4nrTAJJ6PNZZM1/CGT-17208-Project-2?node-id=0-1&t=Ki64MlgAA4Qx5leg-1>

Project Summary

Our project focuses on enhancing the online retail fashion shopping experience through an AI-powered virtual try-on system. The platform addresses common issues such as inconsistent sizing, lack of accurate visualization, and high product return rates by integrating a realistic virtual try-on feature directly within the app interface. AI-based body and facial recognition, shoppers will be able to preview how garments will look on them in picture form, creating a personalized and confident shopping experience.

The unique value proposition is the bridging of the gap between in-store fitting confidence and online convenience. This reduces return rates, boosts purchase confidence, and increases engagement by transforming the shopping experience process into an immersive, user-centered experience.

Target Users

The target audience is online retail fashion shoppers, specifically:

- Individuals frustrated by inconsistent sizing across brands
- Users who want to visualize fit and style before purchasing
- Remote shoppers who lack access to physical retail stores
- Tech-savvy consumers who value convenience and personalization

I. Planning and Thinking

Chosen Planning Activity/Activities

Chosen Stages

P1 Investigate Issues

P2 Assess Feasibility

P8 Explain Design Rationale

Justification

These stages formed the foundation for understanding the real-world frustrations of online shoppers, determining the viability of our idea, and grounding the final design in UX principles. Investigating user pain points revealed the emotional and functional gaps in online retail, and the feasibility stage ensured that the concept was technically and financially sound. The design rationale tied everything together by aligning our solutions with established UX/accessibility principles.

Execution of Chosen Activity

P1 – Investigate Issues / Innovation Opportunities

Key Problems Identified:

What size am I, really? - Inconsistent sizing charts across brands cause uncertainty and anxiety.

How will this look on me? - Static product images fail to show real-life fit, fabric drape, or color accuracy.

The Nightmare of Returns - Ill-fitting products lead to time-consuming, costly returns.

Access for Everyone - Users without access to physical stores need comparable confidence when shopping online.

Our Innovation:

Virtual Fitting Room: Brings realistic garment visualization to users' screens.

Enhanced Personalization: Users can digitally view items on their own body and complexion.

Reduced Return Rates: Fit accuracy lowers the frequency and cost of returns.

Increased Purchase Confidence: Accurate previews make shoppers more decisive and satisfied.

P2 – Assess Feasibility and Cost-Effectiveness

Feasibility Summary:

Our virtual try-on system leverages proven technologies in AI modeling and image rendering. Unlike standalone apps, our approach integrates directly into existing retail websites for seamless scalability.

Cost-Benefit Analysis:

For Shoppers:

- Eliminates uncertainty by showing realistic previews.
- Saves time and money—fewer returns and multiple-item purchases.
- Increases satisfaction and trust in online retail experiences.

For Retailers:

- Financial Gains: Higher conversion rates due to buyer confidence.
- Reduced Costs: Lower return shipping and handling expenses.
- Customer Loyalty: Improved brand trust through innovation and personalization.

P8 – Explain Design Rationale (UX Principles Applied)

Consistency:

- Visual: Consistent header and typography hierarchy across all pages.
- Interaction: Uniform dropdown menus, button shapes, and feedback patterns.
- E-commerce Standards: Familiar interface elements (price visibility, size/color selectors).

User Centricity:

- Personalized avatars for realistic virtual try-on.
- Optional try-on feature to respect user privacy and comfort levels.
- Clear consent modal before photo uploads to build trust and transparency.

Context:

- Clear page headers and back navigation ensure orientation.
- Descriptive instructions guide users through upload and selection processes.
- Feedback elements (“You look great!”) confirm successful completion.

Accessibility:

- High contrast (black-on-white) and scalable text ($\geq 14\text{px}$).
- Text labels for color swatches (e.g., “Rich Sapphire”) to support colorblind users.
- Large tap areas and consistent layout for motor and cognitive accessibility.

II. User Research & Insights

Chosen Research/Analysis Activity/Activities

Chosen Stage(s):

P4 Interviews and Observations

P5 Analyze Data / Task Flow Diagram

Justification:

Direct feedback and observation provided critical insight into user frustrations and mental models of online shopping. Task flow analysis allowed us to translate these insights into efficient user journeys for our application.

Execution of Chosen Activity

P4 – Interviews and Observations

Interview Questions:

- When online shopping, what specific steps do you take to confirm the correct size? Which of those steps feels the most unreliable or frustrating?
- Tell me about a time an online clothing order looked significantly different in person than it did on the website. What was the most critical visual element that the website showed differently?
- Think about the last time you returned a clothing item. At what point in the ordering process did you first have the doubt that you might have to return it? Was it before buying, or only after trying it on?
- If you could adjust the product display to make the clothing look more relevant to you, what is the single most important change you would make to the model or visual presentation?
- If a digital feature could guarantee garment fit before you buy, what would you need to see to eliminate all doubt and feel completely confident in your purchase?

Interview Recording:

https://drive.google.com/file/d/14a9BOpDF23Q_6tEwo69ubgfoStHCud3d/view?usp=drive_link

Observation Analysis:

During informal observation, users frequently rechecked size charts and compared similar items across multiple websites before making a purchase decision. This behavior revealed a general lack of trust in the accuracy of online product displays and sizing information. Users expressed uncertainty about how clothing would fit or appear in real life, often citing inconsistent sizing standards across brands. These insights highlighted the need for features that build confidence

in online shopping experiences. The virtual try-on function was therefore designed to address these concerns by allowing users to visualize fit and appearance more accurately, reducing hesitation and the need to cross-check information on external sites.

System Usability Scale (SUS):

Score=85

The System Usability Scale Standard Version		Strongly Disagree					Strongly Agree				
		1	2	3	4	5					
1	I think that I would like to use this system frequently.										
2	I found the system unnecessarily complex.										
3	I thought the system was easy to use.										
4	I think that I would need the support of a technical person to be able to use this system.										
5	I found the various functions in this system were well integrated.										
6	I thought there was too much inconsistency in this system.										
7	I would imagine that most people would learn to use this system very quickly.										
8	I found the system very awkward to use.										
9	I felt very confident using the system.										
10	I needed to learn a lot of things before I could get going with this system.										

P5 – Analyze Data / Task Flow Diagram

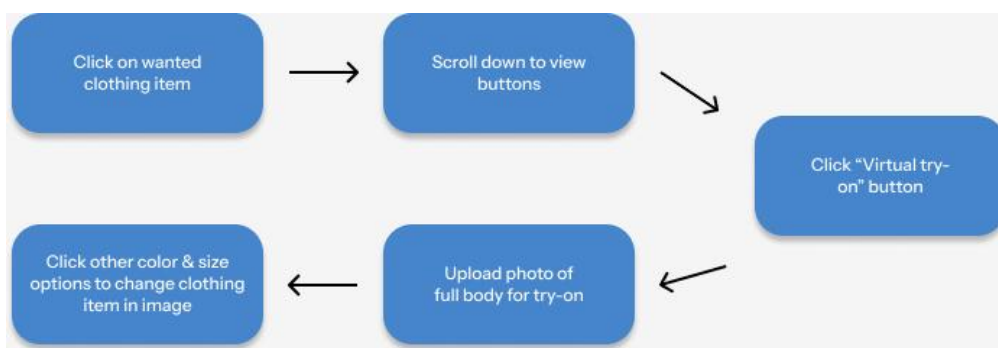
Task Flow Objective:

To map the steps of using the virtual try-on feature from product selection to viewing results, identifying friction points like unclear instructions or upload errors.

Key Insight:

Users needed reassurance and clear feedback at each step — especially before uploading personal photos.

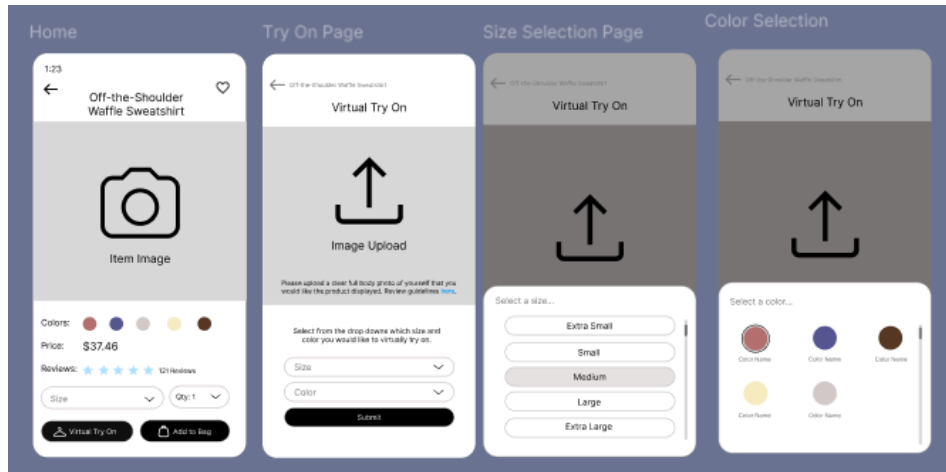
Task Flow Diagram:



III. Prototype and Design

Low-Fidelity Prototyping & Iteration

Low-Fi Evidence: Figure 3 - Low-Fidelity Sketches



Iteration Narrative:

The early low-fidelity prototype focused on mapping the core user flow for the virtual try-on feature. The goal at this stage was to establish how users would move seamlessly from browsing a product to uploading an image for a virtual fitting. Initial sketches defined the placement of key elements such as the product image, “Add to Bag” and “Virtual Try-On” buttons, as well as size and color selection panels. These wireframes helped the team visualize the layout hierarchy and ensure that essential actions were easily discoverable.

During early feedback sessions, users and peers noted that while the layout was clear, the navigation between screens felt slightly disconnected. Specifically, transitions between the “Virtual Try-On,” “Size Selection,” and “Color Selection” pages needed more visual consistency. Based on this feedback, the next iteration prioritized simplifying the navigation flow and strengthening visual continuity across screens. The upload section was redesigned with clearer instructions and improved contrast to ensure better visibility. Button styles and spacing were refined to create a stronger visual hierarchy, helping users quickly identify primary actions such as uploading an image or selecting a size.

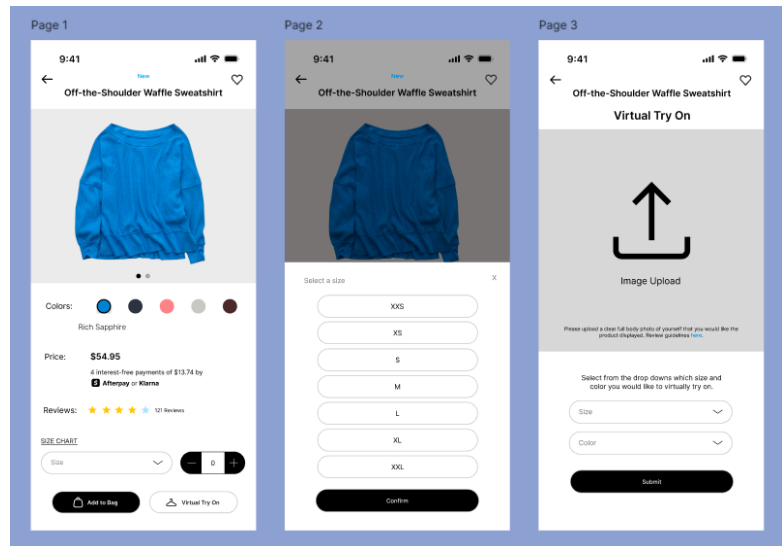
Additional adjustments were made to enhance accessibility and overall usability. This included improving text readability, ensuring sufficient spacing between interactive elements, and maintaining consistent alignment and iconography throughout the interface. These refinements aimed to make the virtual try-on process feel more intuitive and inclusive for a broader range of users.

Overall, the iteration process allowed the design to evolve from a functional wireframe into a more cohesive and user-centered interface. The feedback-driven adjustments not only improved navigation clarity but also reinforced the app's focus on accessibility, ease of use, and an enjoyable shopping experience.

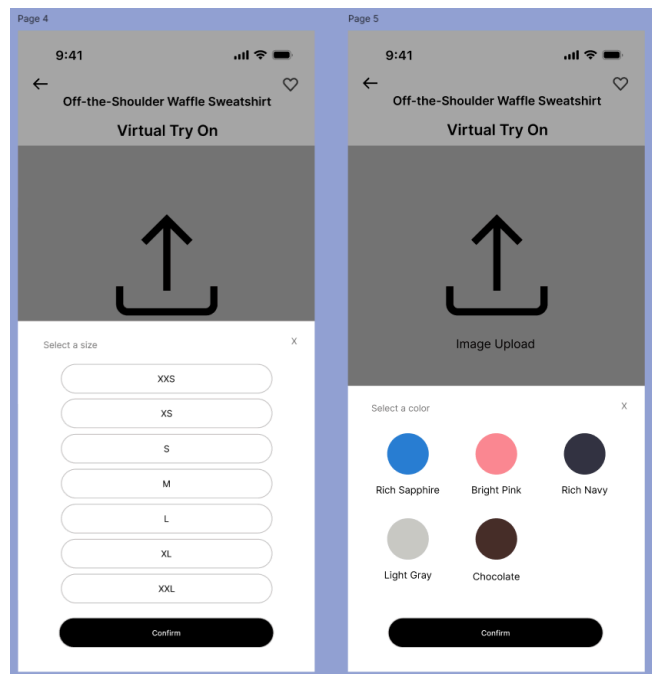
High-Fidelity Visual Design & Polish

Visual Documentation:

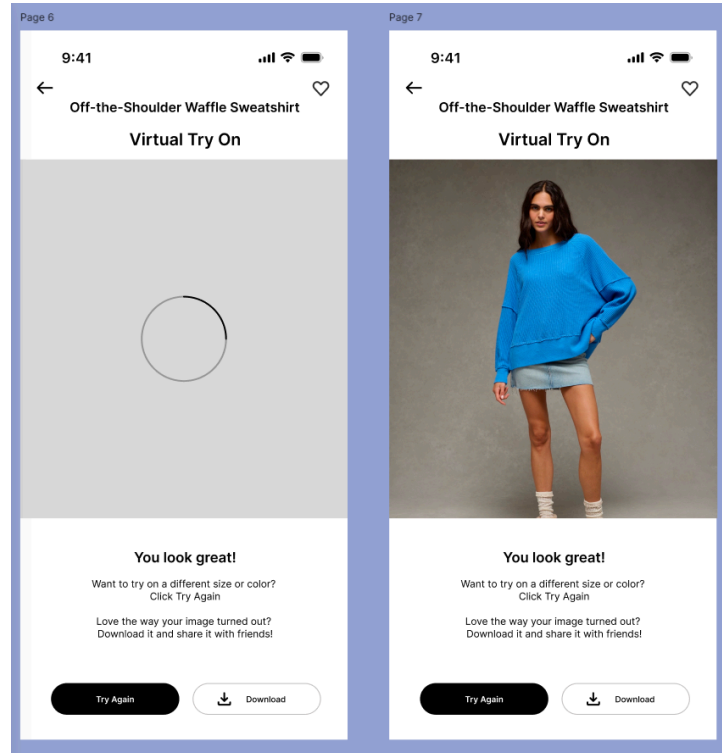
High-fidelity designs (Pages 1–3)



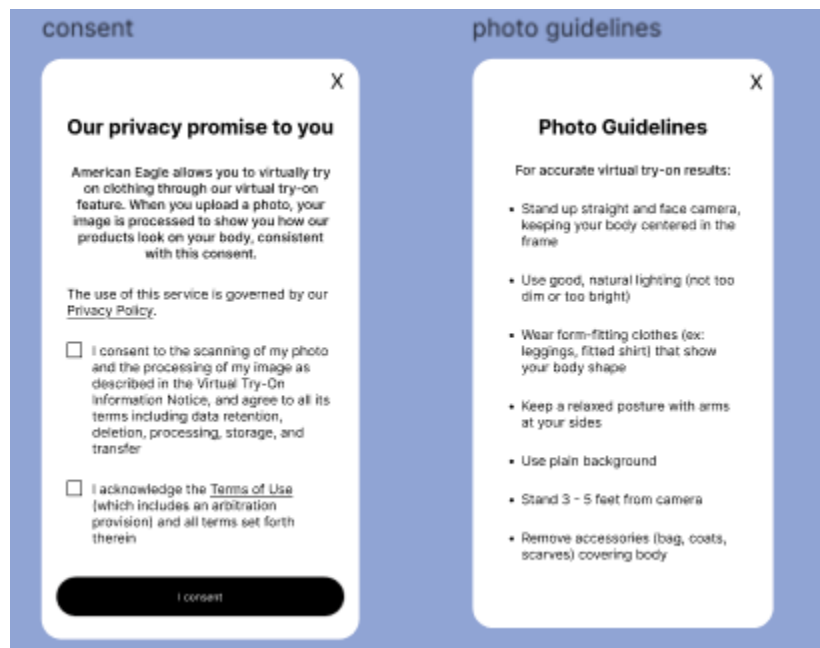
High-fidelity designs (Pages 4–5)



High-fidelity designs (Pages 6–7)



Overlay message designs



Design Rationale:

Color Palette: Simple, minimalistic black-and-white interface for professionalism and contrast

Typography: Clean sans-serif fonts to ensure readability and hierarchy. Meets WCAG 2.1 AA standards (4.5:1 minimum)

Visual Hierarchy: Product image prominence, large actionable buttons, and centered headers to maintain attention flow

Consistency: Repeated header/footer formats unify the browsing experience across pages

High Fidelity Interaction & Functionality

Main Flows:

1. Product Selection - Try-On Activation:
 - The user browses clothing, chooses “Virtual Try-On” or “Add to Bag.”
 - Option empowers both traditional and new-tech users.
2. Consent & Upload Flow:
 - Explicit privacy consent modal ensures user trust.
 - Clear upload instructions reduce confusion and photo errors.
3. Try-On Experience:
 - AI generates accurate visualization of clothing on the user’s body from image.
 - Users can retry, change color/size, or download the rendered image.
4. Completion Feedback:
 - Success message (“You look great!”) reinforces confidence and engagement.

IV. Individual Contribution List

Team Member 1: Shayla Hufford

- Contributions: idea generation, final document formatting, task-flow diagram, and helped with high-fidelity design (general layout, some interactions)

Team Member 2: Kayla

- Contributions: low-fidelity design, interview process and SUS analysis

Team Member 3: Syahirah

- Contributions: research analysis, designed high-fidelity system (layout, working components, most interactions), and investigation during horizon scanning